Variance-Reduction Composition in Conditional Flow Matching: From 2D Experiments to CIFAR-10

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Abstract

Conditional Flow Matching (CFM) trains continuous normalizing flows via simulation-free velocity regression, but the single-sample conditional target induces high gradient variance. Three independent techniques address this: minibatch optimal-transport coupling (BatchOT), importance-weighted marginal velocity targets (StableVM), and antithetic temporal sampling with consistency regularization (TPC). We investigate whether these techniques compose additively through two complementary studies. On 2D synthetic distributions, we confirm that OT-type paths produce straighter trajectories and superior low-NFE robustness. On CIFAR-10, we cross the three axes in a 2^3 factorial ablation: only one of eight cells converged—BatchOT with standard CFM and uniform sampling (FID 18.4). The other seven diverged or oscillated due to dimension-dependent numerical instabilities invisible in 2D. These negative results reveal that StableVM's importance weights scale catastrophically with dimension (d=3072), TPC's consistency penalty causes gradient explosions, and the three axes are not additive. BatchOT alone is the single most impactful variance-reduction technique.

1 Introduction and Background

Flow Matching (Lipman et al., 2023) trains continuous normalizing flows by regressing a neural velocity field $v_\theta(x,t)$ against a target vector field that transports a prior $p_0 = \mathcal{N}(0, I)$ to data p_1. Conditional Flow Matching (CFM) makes this tractable by decomposing the marginal target into conditional terms. For each data point x_1, a conditional probability path $p_t(x|x_1)$ defines a Gaussian interpolation, and the CFM loss regresses against the conditional vector field:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,x_0,x_1} \left\| v_\theta(x_t, t) - u_t(x_t|x_1) \right\|^2 . \tag{1}$$

Under the **OT probability path** (Lipman et al., 2023), the interpolation is linear: $\mu_t(x_1) = t{\cdot}x_1$, $\sigma_t = 1 - (1 - \sigma_{\min})t$ with $\sigma_{\min} = 10^{-4}$, yielding:

$$u_t(x|x_1) = \frac{x_1 - (1 - \sigma_{\min})x}{1 - (1 - \sigma_{\min})t}. \tag{2}$$

While unbiased, the single-sample conditional target has high variance. Three techniques address this along independent axes:

BatchOT coupling. Tong et al. (2024) replace independent noise–data pairing with minibatch optimal transport: within each batch, Sinkhorn's algorithm (Cuturi, 2013) finds a low-cost assignment between noise and data, producing shorter, straighter interpolation paths.

StableVM objective. For each x_t, K=16 references $\{x_1^{(k)}\}$ are drawn from a memory bank. Importance weights $w_k = \text{softmax}_k(\log p_t(x_t|x_1^{(k)}))$ yield a multi-reference target $u_t^{\text{VM}} = \sum_k w_k u_t(x_t|x_1^{(k)})$. The log-probability under the OT path is:

$$\log p_t(x|x_1) = -\tfrac{d}{2}\log(2\pi) - d\log\sigma_t - \tfrac{1}{2\sigma_t^2}\|x - \mu_t(x_1)\|^2. \tag{3}$$

TPC estimator. Antithetic pairs $t_1 \sim \mathcal{U}[0, 0.5]$, $t_2 = 1 - t_1$ replace uniform sampling, with a consistency penalty:

$$\mathcal{L}_{\text{TPC}} = \mathcal{L}(t_1) + \mathcal{L}(t_2) + \lambda\|v_\theta(x_t^{(1)}, t_1) - v_\theta(x_t^{(2)}, t_2)\|^2, \quad \lambda = 0.1. \tag{4}$$

Research questions. Each technique works in isolation, but their composition is unexplored. We ask: **(RQ1)** Are the three axes additive? **(RQ2)** Which single axis helps most? **(RQ3)** Do 2D geometric insights transfer to CIFAR-10?

2 Experimental Setup

Study 1: 2D synthetic distributions. Four flow-matching variants (OT-CFM, VP-CFM, Target-CFM, SB-CFM) are compared on `checkerboard`, `two_moons`, and `gaussian_mixture`. The velocity field is a time-conditioned MLP (3 layers, 256 units, SiLU). Quality is measured via SWD and MMD at NFE=64, with additional trajectory curvature and NFE-robustness diagnostics.

Study 2: CIFAR-10 factorial ablation. The three variance-reduction axes are crossed in a full 2^3 design (Table 1). All 8 cells share a U-Net backbone (∼9M parameters, base channels 64, attention at 16×16), trained with Adam (lr=10^{-4}), EMA decay 0.9999, fp16 mixed precision, batch size 256, for 80k steps on CIFAR-10. Training ran on AMD MI210 GPUs. FID (Heusel et al., 2017) is evaluated at NFE $\in \{5, 10, 20, 40, 100\}$ with Euler, midpoint, and RK4 solvers (15 evaluations per cell). Cells that diverged to NaN used a pre-divergence checkpoint (step 10k) for evaluation.

Table 1: 2^3 factorial design. Binary: bit 2 = Coupling, bit 1 = Objective, bit 0 = Estimator.

Cell	Code	Coupling	Objective	Estimator
0	000	Independent	CFM	Uniform
1	001	Independent	CFM	TPC
2	010	Independent	StableVM	Uniform
3	011	Independent	StableVM	TPC
4	100	BatchOT	CFM	Uniform
5	101	BatchOT	CFM	TPC
6	110	BatchOT	StableVM	Uniform
7	111	BatchOT	StableVM	TPC

3 Study 1: 2D Experiments

Across three 2D datasets, two clear winners emerge. **SB-CFM** achieves the best endpoint quality on all three datasets (SWD@64 and MMD@64). **OT-CFM** produces the straightest trajectories (lowest curvature) on all three datasets and wins low-NFE robustness on two of three. At NFE≤8, OT and SB variants achieve SWD ∼0.022, while VP and Target reach ∼0.040—roughly 47% worse. Table 2 summarizes the multi-dataset comparison.

Table 2: Win counts across 3 datasets and 5 metrics (out of 3 possible wins each).

Variant	SWD@64	MMD@64	Low-NFE	Curvature	AUC
OT-CFM	0	0	**2**	**3**	0
VP-CFM	0	0	0	0	0
Target-CFM	0	0	0	0	1
SB-CFM	**3**	**3**	1	0	**2**

Crucially, **all four variants work flawlessly in 2D**—no instabilities, no divergence, smooth convergence across all datasets. This will prove misleading: the 2D setting masks numerical issues that become catastrophic in high dimensions (Section 5).

4 Study 2: CIFAR-10 Ablation

4.1 Training Outcomes

Table 3 presents the central result: only 1 of 8 cells converged.

Table 3: Training outcomes for all 8 cells. Best FID is the minimum across 15 solver/NFE combinations.

Cell	Coupling	Objective	Estimator	Status	Best FID	Ckpt
000	Indep.	CFM	Uniform	NaN	679.2	10k
001	Indep.	CFM	TPC	NaN	419.8	10k
010	Indep.	StableVM	Uniform	Unstable	256.8	final
011	Indep.	StableVM	TPC	Unstable	197.9	final
100	**BatchOT**	**CFM**	**Uniform**	**OK**	**18.4**	**final**
101	BatchOT	CFM	TPC	NaN	679.2	10k
110	BatchOT	StableVM	Uniform	Unstable	211.4	final
111	BatchOT	StableVM	TPC	Unstable	273.6	final

Three patterns are immediate: (i) BatchOT is necessary but not sufficient—cell_100 converged, but 101/110/111 did not; (ii) TPC causes NaN in every configuration (001, 101); (iii) StableVM causes extreme loss oscillation ($\sim$40 to $\sim$22,000) in all four cells where it appears (010, 011, 110, 111).

4.2 Cell_100: The Only Successful Model

Figure 1 shows the FID–NFE trade-off for cell_100. Training loss decreased smoothly from 0.54 to 0.14 over 80k steps. Midpoint is the most efficient solver, reaching FID 20.4 at NFE=40—matching Euler at NFE=100, a 2.5$\times$ efficiency gain consistent with Study 1's trajectory-straightness findings. RK4 achieves the best absolute FID of 18.4 at NFE=100.

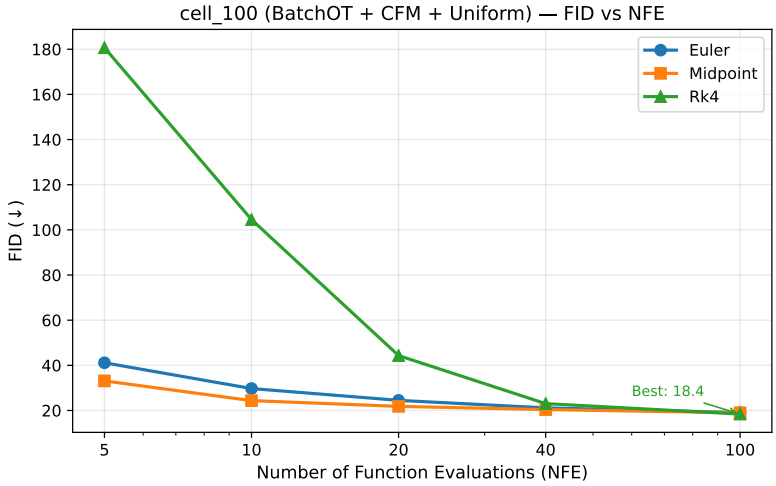

Figure 1: FID vs. NFE for cell_100 (BatchOT + CFM + Uniform). Midpoint dominates at low NFE; RK4 wins at high NFE.

5 Discussion and Conclusion

5.1 Failure Analysis

Baseline divergence (cell_000). The OT path denominator $1-(1-\sigma_{\min})t$ reaches 10^{-4} near t=1, producing velocity magnitudes $\sim 10^4$. Under fp16, this overflows after ∼18,700 steps. BatchOT prevents this: by minimizing transport cost, it keeps $\|x_1 - x_0\|$ small, so both the numerator and denominator of Eq. 2 remain moderate. BatchOT is not just a variance reducer—it is a *numerical stabilizer* for the OT path.

StableVM collapse (cells 010/011/110/111). The log-probability (Eq. 3) contains the term $-d \log \sigma_t$. For CIFAR-10 ($d = 3{\times}32{\times}32 = 3072$) near t=1:

$$-d \log \sigma_t \approx -3072 \times \log(10^{-4}) \approx 28{,}300.$$

This enormous value causes the softmax over importance weights to collapse onto a single reference sample, producing a noisy, numerically unstable target. In 2D (d=2), the same term yields only ∼18.4, explaining why StableVM works perfectly in Study 1 but oscillates wildly on CIFAR-10.

TPC gradient explosion (cells 001/101). The consistency penalty $\|v_\theta(x_t, t_1) - v_\theta(x_t, t_2)\|^2$ compares velocities at $t_1 \approx 0$ and $t_2 \approx 1$. Near t=1, velocity magnitudes are $\sim 1/\sigma_{\min} = 10^4$; near t=0, they are $O(1)$. The penalty gradient is dominated by the large-t term, causing gradient explosions that trigger NaN at steps 14–17k.

5.2 Answers to Research Questions

RQ1: The three axes are *not* additive—combining BatchOT with StableVM or TPC *destroys* the convergence that BatchOT alone provides. **RQ2:** BatchOT is unambiguously the most impactful single axis, consistent with Tong et al. (2024). **RQ3:** Geometric insights transfer (OT paths are straighter, solvers are more efficient), but stability insights do not: techniques safe in 2D fail catastrophically at d=3072.

5.3 Limitations

All cells used a single seed, fast mode (9M parameters, 80k steps), and fp16 mixed precision without gradient clipping. Gradient clipping and fp32 training might rescue the baseline (cell_000), but the StableVM and TPC instabilities are structural—they stem from dimension-dependent scaling, not precision. The two studies use different codebases (MLP vs. U-Net), so direct numerical comparisons between them are not meaningful.

5.4 Conclusion

BatchOT coupling is validated as the key variance-reduction ingredient for CFM. StableVM and TPC require dimension-aware redesigns before deployment on image data: normalizing importance weights by $1/d$, restricting TPC pairs to $t \in [0.1, 0.9]$, or detaching the consistency penalty from the gradient. More broadly, variance-reduction techniques must be validated not only for their statistical properties but for their numerical stability in the target dimension.

References

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